

Jay H. Jivandas

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EDUCATION

The Ohio State University, College of Engineering (3.6 GPA) **Columbus, OH**
Bachelor of Science in Computer Science and Engineering, AI Specialization
Minors: Statistics and Business, Engineering Honors *May 2026*

- Honors Research Distinction; Dean's List (4/6 semesters)

PROFESSIONAL EXPERIENCE

SpiceBox.AI **Dubai, UAE**
AI/ML Development Intern *May 2025 – Present*

- Building foundational components for an AI-powered law firm, aligning with the UAE's vision for an AI-driven city
- Developed tools to automate legal workflows for SMEs, reducing overhead by over 50%

Lifestyle Communities **Columbus, OH**
Application Development Intern *May 2024 – July 2024*

- Created reusable Laravel/PHP/JS components, saving 15+ dev hours per project
- Rebuilt coupon management system with senior developers to streamline discount generation, distribution, and tracking

BMB Group **Dubai, UAE**
Full Stack Web Development Intern *June 2023 – Aug 2023*

- Rebuilt maintenance app (PHP/MySQL/JS) used by 1,000+ employees, improving repair efficiency by 20%

RESEARCH EXPERIENCE

Imageomics Institute + Infectious Diseases Institute, The Ohio State University **Columbus, OH**
Undergraduate Research Assistant *May 2025 – Present*

- Processed and curated 2,000+ tick images with metadata using PyBioCLIP, pandas, and PIL for classification and feature extraction
- Preparing for model fine-tuning (SVM and other methods) with scikit-learn, incorporating dataset expansion and performance evaluation via confusion matrices

AgSensing Data Analytics for Agriculture, The Ohio State University **Columbus, OH**
Undergraduate Research Assistant *Jan 2025 – Present*

- Preparing large-scale agricultural datasets for ML models to predict optimal nitrogen rates
- Co-authoring a peer-reviewed research paper on methodology and findings

Imageomics Institute, The Ohio State University **Columbus, OH**
Undergraduate Research Assistant *Aug 2024 – Dec 2024*

- Evaluated deep learning and computer vision models (SAM, YOLO, TensorFlow, PyTorch) on 1,000+ butterfly images, applying bounding boxes for species identification and classifications
- Collaborated with PhD students to improve model accuracy and refine algorithms

Civil, Environmental and Geodetic Engineering Department, The Ohio State University **Columbus, OH**
Undergraduate Research Assistant *Aug 2023 – April 2024*

- Collected and processed raw flux data from two ecosystem measurement sites using MATLAB

PROJECTS

S&P 500 Trend Prediction Model

- Built a machine learning model to forecast S&P 500 trends using Yahoo Finance data and engineered market indicators
- Achieved ~60% accuracy through iterative tuning, feature selection, and historical backtesting

OSU Engineering Org Assistant

- Built a chat-based assistant using Llama API to help students find engineering orgs based on interests and queries

Weather Assistant App

- Developed a React Native weather assistant app with OpenWeather API, offering live forecasts, clothing tips, and daily weather-based alerts
- Integrated location-based data, hourly UV index, and rain notifications with a responsive, mobile-friendly UI

QUALIFICATIONS

- Programming Languages:** Python, Java, C, C++, SQL, PHP, JavaScript, HTML/CSS, MATLAB, R, Ruby, Bash
- Frameworks & Libraries:** PyTorch, TensorFlow, scikit-learn, pandas, NumPy, PIL, YOLO, SAM, React Native, Hugging Face, PyBioCLIP
- Tools & Platforms:** Git, Linux, Tableau, SolidWorks
- Relevant Coursework:** Machine Learning, Computer Vision, Statistical Modeling for Discovery I (STAT 3301), Data Structures and Algorithms, Software Development, Database Systems, Intro to AI, Linear Algebra, Calculus I & II, Discrete Structures, Foundations of Higher Mathematics

LEADERSHIP & ACTIVITIES

OSU Club Water Polo, Student Athlete *Aug 2024 – Present*
OSU Undergraduate Honors Committee, Student member (1 of 6) *Aug 2023 – May 2025*
FIRST Alumni and Robotics, Student Mentor for a group of high school students *Oct 2023 – April 2024*